

Zero to One

Notes on Startups or
How to Build the Future

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The Big “So What”



How did companies like Google, Paypal, Amazon and Telsa become so successful, and how can you get there as a startup? In this book, Thiel – a billionaire entrepreneur and investor who co-founded PayPal and Palantir, and invested in hundreds of startups including Facebook and SpaceX - shares his insights about business and entrepreneurship. He shares why much of what we believe in may be flawed, and how we can think about and create truly valuable businesses that shape a better world.

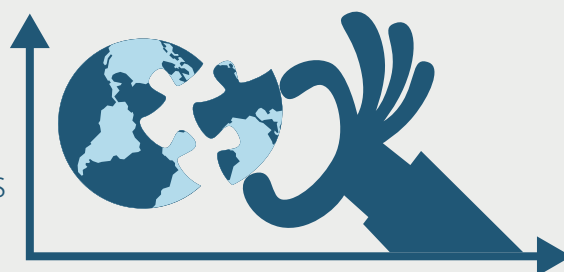
Introduction



Changing the Future - Zero to One

0 to 1

- Doing new things
- Vertical /intensive progress
- Driven by Technology



1 to n

- Copying things that work
- Horizontal/ extensive progress
- Driven by Globalization

Progress can take 2 forms:

Horizontal (extensive) progress, which comes from copying things that work, e.g. when you improve a typewriter and sell it globally. It brings us from **1 to n**.

Vertical (intensive) progress, which comes from doing new things that haven't been done before, e.g. when you move beyond typewriters to invent a word processor. It brings us from **0 to 1**.

KEY QUOTES

“Every moment in business happens only once. The next Bill Gates will not build an operating system. The next Larry Page or Sergey Brin won't make a search engine. And the next Mark Zuckerberg won't create a social network. If you are copying these guys, you aren't learning from them.”

“The paradox of teaching entrepreneurship is that such a formula necessarily cannot exist; because every innovation is new and unique.”

“This book is about the questions you must ask and answer to succeed in the business of doing new things... an exercise in thinking.”

The most successful businesses in the world – such as Apple, Facebook, Google – achieved success not by copying others, but by creating something fundamentally new. They moved from 0 to 1. Globalization is at the heart of horizontal progress, while technology is at the heart of vertical progress. And the good news is, both can coexist.

The most valuable companies became so because they uncovered truths behind widely-held (but incorrect) beliefs, hence fundamentally changing our world. Thiel calls these “secrets”. Every multi-billion dollar company was once a startup, a small group of like-minded people who fervently pursued alternative truths to build a different future.

Revisiting the 1990s Dot-Com Mania

In the book, Thiel gives a quick recap of the dismal global settings in the 1990s, the rise of the dot-com mania, and the burst of the dot-com bubble in 2000. Here are 2 important takeaways:



First, although the dot-com mania was short-lived, it was a reflection of growing concerns that the “old economy” wasn’t working. The internet held promise of solutions for a globalized world, and indeed, technology and the internet have helped to tackle many challenges of globalization today.



Second, much of what we “know” is shaped by circumstances and reactions to past mistakes, and may not be true at all. Thiel showed how the business “wisdom” in the post dot-com era was actually flawed. We’ll now expand on some of Thiel’s business insights and ideas in the book.

Zero to One Business Insights



Thiel challenges several market and business wisdoms that are widely accepted. This segment provides an overview of the key ideas.

KEY QUOTES

“What makes the future distinctive and important isn’t that it hasn’t happened yet, but rather that it will be a time when the world looks different from today.”

“Positively defined, a startup is the largest group of people you can convince of a plan to build a different future.”

“The first step to thinking clearly is to question what we think we know about the past.”

“The most contrarian thing of all is not to oppose the crowd but to think for yourself.”

“Ask yourself: how much of what you know about business is shaped by mistaken reactions to past mistakes?”



Most of us grew up learning that competition is good and essential. Thiel suggests the opposite – that **competition is destructive while a “creative monopoly” is the way of the future**. Thiel defines a creative monopoly as one that has achieved this status because it does such a great job that there’re no viable substitutes.



It’s as important to create value as it is to retain some of that value. Businesses fighting cut-throat competition are so preoccupied with survival that they can’t focus on anything else. For instance, airlines create tremendous value through air travel, but are barely profitable. On the contrary, a creative monopoly like Google has not had viable substitutes - including Yahoo! and Microsoft – since the early 2000s, allowing it to retain some 20% of its revenues. It has resources to care about its stakeholders, invest in long term research and innovation, and make a greater impact on the world.



It’s true that monopolies can be bad if nothing changes, and customers are forced to pay high prices with limited or no choices. But **our world is a dynamic one, with better monopoly businesses replacing incumbents to offer more consumer choice and societal improvement**. In any case, our perception of the level of competition is usually inaccurate. Monopolists lie and exaggerate the level of (non-existent) competition to protect themselves from being attacked, while non-monopolists lie and understate the level of competition by positioning themselves as the leader of their narrowly-defined (sometimes non-existent) niche.



Competition can shift the attention away from value-creation and productivity to beating our rivals or even scoring personal victories.

- Obviously, in business, we’ll sometimes have to go all out and fight to win. But **in many cases, competition is unnecessary**. For example,

KEY QUOTES

“War is costly business. Rivalry causes us to overemphasize old opportunities and slavishly copy what has worked in the past.”

“In business, money is either an important thing or it is everything...Only one thing can allow a business to transcend the daily brute struggle for survival: monopoly profits.”

“All happy companies are different: each one earns a monopoly by solving a unique problem. All failed companies are the same: they failed to escape competition.”

“If the tendency of monopoly businesses were to hold back progress, they would be dangerous... But the history of progress is a history of better monopoly businesses replacing incumbents.”

“It’s competition, not business, that is like war: allegedly necessary, supposedly valiant, but ultimately destructive.”

Microsoft and Google are both great companies with different creations, and no obvious reason to fight. Yet, they were locked in fierce competition for years. Apple overtook them in January 2013 with a market capitalization of \$500b (vs their combined capitalization of \$467b), when just 3 years before, each company was individually more valuable than Apple. The same pattern is seen from credit card readers to online pet stores.

- By contrast, PayPal was also locked in fierce competition with X.com in the late 1990s. However, realizing the risk from the rapidly inflating tech bubble, the 2 companies decided on a 50-50 merger in 2000, which allowed them to ride out the dot-com crash that came soon after.



So why do we compete? Because **the ideology of competition is so entrenched in our society that we end up being trapped in it.** It starts with our education system, which is built on grades and ranking students against one another. When we enter the workplace or business world, we are constantly trying to beat our competition and compete with our peers for career advancement.

In short, Thiel believes that competition destructs rather than creates value, and advocates the concept of monopoly capitalism.

The Power Law Of Venture Capital



Venture capitalists (VCs) try to find, fund, and profit from promising early-stage companies. They make money when the companies in their portfolio go public or get bought by larger companies. Most venture funds last for 10-years, as that's usually how long successful companies take to grow and "exit."

KEY QUOTES

"We preach competition, internalize its necessity, and enact its commandments; and as a result, we trap ourselves within it—even though the more we compete, the less we gain."

"There is no middle ground: either don't throw any punches, or strike hard and end it quickly."

"Winning is better than losing, but everybody loses when the war isn't one worth fighting."

"Amid all the human drama, people lose sight of what matters and focus on their rivals instead."

"The biggest secret in venture capital is that the best investment in a successful fund equals or outperforms the entire rest of the fund combined."

"VCs must find the handful of companies that will successfully go from 0 to 1 and then back them with every resource."



Unfortunately, the reality is that most start-ups fail (usually soon after they start), and a few companies will drastically outperform all others. **The one best investment in a successful fund will outdo the rest of the companies combined.** A good VC should only invest in companies that can potentially return the value of the entire fund. Obviously, this is hard to predict, so a VC will still need a “portfolio”. But, **every single company in a good portfolio must have the potential to go from 0 to 1 and succeed at large scale.**



This power law doesn't only apply to VCs. All of us are investors - entrepreneurs invest time and money in their startups, and employees invest time in their career. **The question to ask is: will what you are doing be valuable decades from now?**

Uncovering “Secrets”



The most valuable companies find and build their business upon “secrets”, i.e. **what the world believes in that isn't true.** For example, Airbnb and Uber saw untapped supply and unmet demand in lodging and transportation, and created businesses to address them.

- **It starts with belief.** Most people act as if all the world's biggest problems have already been solved, there're no more opportunities left to find, hence they don't even try. But, so long as there are injustice and inefficiencies in this world, there are secrets to be found. Start by asking: “What valuable company is nobody building?” This will lead you to address hard but potentially world-changing questions.
- **Finding secrets.** **There are two kinds of secrets:**
 1. Secrets of nature: These are undiscovered aspects of the physical or natural world.
 2. Secrets about people. These are things that people hide, or don't know about themselves.

The best place to look for secrets is where no one else is looking. **Ask if are there any areas that matter, but haven't been standardized or institutionalized,** e.g. Physics is

KEY QUOTES

“Once you think that you're playing the lottery, you've already psychologically prepared yourself to lose.”

“You can't find secrets without looking for them.”

“If you think something hard is impossible, you'll never even start trying to achieve it. Belief in secrets is an effective truth.”

taught systematically in schools, whereas nutrition (which affects everyone) is not.

- **What to do with Secrets.** Generally, your secret or viewpoint is likely to be opposite of what the masses believe. **When you share your secret, you are actually enlisting fellow conspirators, so do it discreetly.**

Last Mover Advantages



Your business value is the sum of all the money it'll make in the future, discounted to the present worth. Being the first mover in a market can give you a headstart, but **what matters is long-term cashflows, and it's useless being the first-mover if you get dethroned later.**



It's better to be the last mover – study the end-game, dominate a small niche and scale up to eventually develop and enjoy years of monopoly profits.



However, for a company to become valuable, it must be durable. **Many entrepreneurs are so focused on short-term growth that they fail to consider if their business will still be around 10 years from now.**

Let's look at some of the questions and foundations to help you think about building a sustainable, monopoly business.

Building a Monopoly



Characteristics of Monopolies



Most successful monopolies share some of these characteristics:



Proprietary Technology (e.g. Google's search algorithms) can be a massive advantage because it makes your product hard or impossible to replicate. To have a real monopolistic advantage, **your technology must be at least 10 times better than any substitute** in some key aspect(s). To achieve that, you can either create something

KEY QUOTES

"A great company is a conspiracy to change the world; when you share your secret, the recipient becomes a fellow conspirator."

"As a rule of thumb, proprietary technology must be at least 10 times better than its closest substitute in some important dimension to lead to a real monopolistic advantage."

completely new, or fundamentally transform an existing solution. For example, PayPal enabled buyers to pay on eBay straight after an auction (instead of mailing a check which took 7-10 days) and Amazon offered at least 10x as many books as any other bookstore.



Network Effects make a product more useful as more people use it, e.g. the more of your friends are on Facebook, the more you'd want to use it. Yet, such businesses must still be valuable to their initial users and small markets.

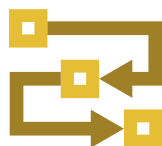


Economies of Scale spread out business fixed costs (e.g. R&D, management, office space) over more users/ sales, such that the marginal cost of producing an extra unit is negligible. When designing your startup, build in the potential to scale up.



Branding is key to any monopoly, but it must be backed by substance. One of the strongest tech brands is Apple – its sleek minimalist design, consumer experience, advertising campaigns, and price positioning are backed by a suite of proprietary technologies, economies of scale and strong network of developers and users.

The Process of Building a Monopoly



As a small start-up, how can you put together the 4 elements above (brand, scale, network effects, and technology) to form your eventual monopoly?



1. Start Small and Monopolize

Every startup is small initially, yet a monopoly must dominate a large share of its market. Hence, don't try to go for 1% of a \$100 billion market. As a startup, your ideal target market is a small, specific group of people concentrated together and served by few or no competitors.

However, make sure the market actually exists. PayPal's first product allowed PalmPilot

KEY QUOTES

"It's easier to dominate a small market than a large one. If you think your initial market might be too big, it almost certainly is."

"It was much easier to reach a few thousand people who really needed our product than to try to compete for the attention of millions of scattered individuals."

users to beam money to each other; but the millions of PalmPilot users were scattered geographically, had little in common and didn't need to beam money to one another, so PayPal had effectively no customers. When they switched their focus to eBay auctions (with only a few thousand high-volume "PowerSellers"), they quickly secured 25% of the market within 3 months.



2. Scale Up Progressively.

Once you've create and dominated a niche, you can expand into related and slightly broader markets gradually and sequentially. Amazon wanted to dominate online retail, but started only with books (which had similar sizes and characteristics despite millions of titles). Then, it expanded into CDs, videos, and software, and gradually added categories until it became the world's general store. eBay launched its auction marketplace with intense interest groups (e.g. Beanie Baby obsessives), then expanded to other small-time hobbyists until it became a full-fledged marketplace for people trading anything online.



3. Don't Disrupt.

"Disruption" means using new technology to introduce a low-end product at low prices, then improving the product over time to eventually overtake the incumbents, e.g. PCs disrupting mainframe computers, then mobile devices disrupting PCs. However, disruption is not about taking the big boys head on – startups that try to do so end up in fights that they can't win. Instead, focus on creating something really new, then expand it gradually to adjacent markets, avoiding competition as much as possible.

Building Strong Foundations



Some foundations cannot be undone, so you must start the company on the right foot:

KEY QUOTES

"Don't disrupt: avoid competition as much as possible."

"Choosing a co-founder is like getting married, and founder conflict is just as ugly as divorce."



Synergy. The right people makes all the difference:



The Right Partner(s): Choosing the right partner for your business is like choosing the right person to marry. Besides technical abilities and complementary skills, the founders should know each other well and have worked well together.



Ownership, Possession, and Control. To minimize misalignment, distinguish between Ownership (who owns the company), Possession (who runs the company), and Control (who officially governs the company). For most startups, the founders usually have both ownership and possession, but conflicts can arise with the board members (who are supposed to exercise control). Thiel recommends that you select every board member carefully, and keep the board small (ideally 3 and maximum 5 members, unless you are a public company). This facilitates communication, agreement and effective control.



Beyond professionalism. To build a long-term business, the people in your company must be not just talented, but actually enjoy working together. Transactional relationships (however good the resumes) won't work.



Long Term Value & Alignment



On the bus or off the bus. Thiel advocates that everyone in your company should either own stock options or be engaged full-time (with the exception of positions like lawyers and accountants). Consultants and part-timers have no vested interest to create long-term value for your company.

KEY QUOTES

"Founders should share a prehistory before they start a company together - otherwise they're just rolling dice."

"In the boardroom, less is more. If you want an effective board, keep it small."

"Since time is your most valuable asset, it's odd to spend it working with people who don't envision any long-term future together."

If you're deciding whether to bring someone on board, the decision is binary...you're either on the bus or off the bus.



Cash vs vested interest. To get commitment, you must compensate people properly. But, **cash compensation fosters short-term thinking** - employees take value from the company instead of create long-term value. Instead of a high cash salary, consider giving outcome-based cash bonuses, offer equity (based on people's talents, responsibilities and opportunity costs) and limit the CEO's annual salary to \$150,000 (so he'll focus on increasing the company's value and set a good example).



Build a Tribe. Startups have tight resources and small teams. To get the most from them:



Build a cult; Do 1 thing. **Hire people who share your obsession** - it's much better to have a team that shares a common world view and is fiercely-devoted to your mission, than a group of lukewarm professionals. Ask, "why should the 20th employee join your company?", to **address why people should join you and who they will work with.** To provide clear responsibility and reduce conflict, match talents with tasks such that **every person in your company is just responsible for just one thing.**



Extend the founding. You can't reverse that moment when your company is founded and rules are set from scratch. However, you can **build a culture of openness and ongoing invention**, so you can re-create the company over and over again.

- **Sales & Distribution.** No matter how good your product, you need to sell it. The best sales are those that are "hidden", i.e. you don't realize you've been sold to.

KEY QUOTES

"Any kind of cash is more about the present than the future."

"People at a successful startup are fanatically right about something those outside it have missed."

"Job assignments aren't just about the relationships between workers and tasks; they're also about relationships between employees."

"(The founding moment) lasts as long as a company is creating new things, and it ends when creation stops."

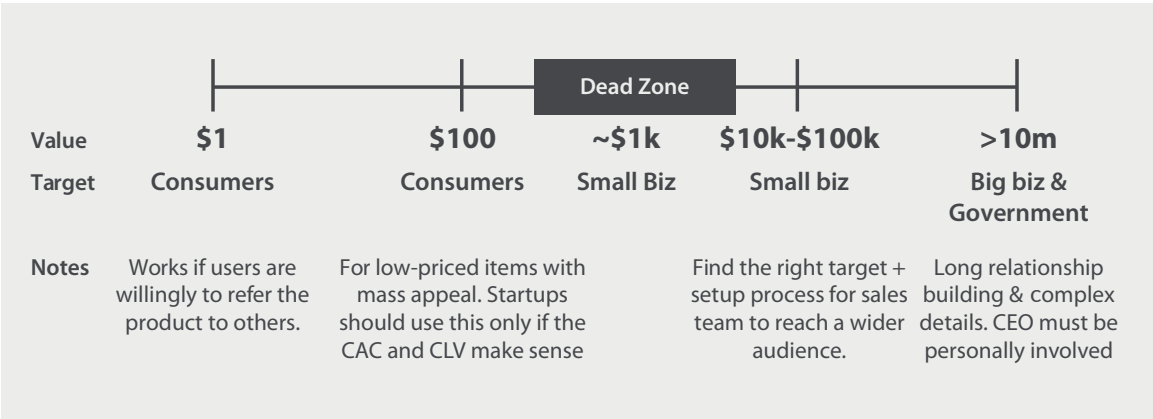
"Customers will not come just because you build it. You have to make that happen, and it's harder than it looks."

1. Apply 2 rules of thumb for effective distribution:

• Rule #1:



- Rule #2: The higher the price of your product, the more you have to spend to make a sale. Thiel recommends sales and marketing approaches for different product categories, summarized here:



Of these methods above, one of the methods will likely be the most effective for your business. It's better to establish 1 sales channel that works rather than spread your resources over many that don't.

2. Sell to non-consumers: Besides selling to customers, you must also sell your company to employees, investors and the media. Media exposure creates publicity for your products and helps you to attract investors and employees.

KEY QUOTES

"If you've invented something new but you haven't invented an effective way to sell it, you have a bad business—no matter how good the product."

"Superior sales and distribution by itself can create a monopoly, even with no product differentiation. The converse is not true."

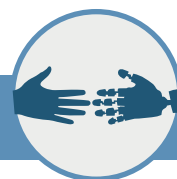
"If you can get just one distribution channel to work, you have a great business. If you try for several but don't nail one, you're finished."

"Everybody sells...Look around. If you don't see any salespeople, you're the salesperson."

Putting it Together

Since the start of the 21st century, clean technology has become the "next big thing", yet billions of dollars have been lost in this industry. Using 7 questions to integrate the points above, Thiel shares why Tesla succeeded where most cleantech companies failed. To secure massive success, you need to address most, if not all, of these 7 questions:

1. Engineering	Can you create breakthrough technology instead of incremental improvements?
2. Timing	Is now the right time to start your particular business?
3. Monopoly	Are you starting with a big share of a small market?
4. People	Do you have the right team?
5. Distribution	Do you have a way to not just create but deliver your product?
6. Durability	Will your market position be defensible 10 and 20 years into the future?
7. Secret	Have you identified a unique opportunity that others don't see?



Man and Machine

With computing capabilities growing exponentially, many people are predicting (and fearing) that computers will replace humans in the coming decades. **Thiel argues that computers are complements, not substitutes to humans, and the most valuable businesses will be those who empower people in a computer age.**

- Unlike people, who compete to supply their labour/ talent, and also demand for the same resources like food, shelter, and better lifestyle, computers compete for nothing.
- People have “intentionality”. We make plans and can exercise judgement. Computers are great at data processing, but they can't exercise basic judgement, much less those in complex situations.

Hence, we shouldn't compete with computers because they are different from humans - they are merely tools. **The key to the future is to create businesses where the human-computer combination could achieve much more than either could alone.**

KEY QUOTES

“The most valuable businesses of coming decades will be built by entrepreneurs who seek to empower people rather than try to make them obsolete.”

“Humans and computers are not just more or less powerful than each other—they're categorically different.”

“Big data is usually dumb data. Computers can find patterns that elude humans, but they don't know how to compare patterns from different sources or how to interpret complex behaviors. Actionable insights can only come from a human analyst.”

“As we find new ways to use computers, they won't just get better at the kinds of things people already do; they'll help us to do what was previously unimaginable.”

For example, PayPal was initially losing millions each month due to credit card fraud. Humans couldn't cope with the volume of transactions, and automated computer-checks were outsmarted by the fraudsters. PayPal solved the problem with a hybrid approach - the computers flag out suspicious transactions, and human analysts apply their judgement for decision-making. Likewise, LinkedIn didn't try to replace recruiters with technology. Instead, they help professionals to build their personal brands, and empower recruiters to search and filter job candidates using technology.

Business founders play a crucial role for the success of their startups, not because of their personal brilliance and produce valuable work, but because they can bring out the best from everyone in their company.

KEY QUOTES

"Founders are important not because they are the only ones whose work has value, but rather because a great founder can bring out the best work from everybody at his company."